

XSpace OS

User Manual

By Whitespace

Members:

Guarin, Nicolete Rein

Legaspo, Jed Lordy

Pagaran, Niño Christian

Solis, Glen

Table of Contents

[I. Introduction and Scope](#)

[II. Requirements and Project Setup](#)

[III. Build and Boot Procedures](#)

[IV. Desktop and Application Navigation](#)

[V. File Editing Workflow](#)

[VI. Allocation Table and File System Model](#)

[VII. Keyboard Reference](#)

[VIII. Troubleshooting](#)

[IX. Demonstration Flow](#)

I. Introduction and Scope

XSpace OS is a small Rust operating system kernel that boots directly in a virtual machine. It uses VGA text mode for screen output and a polled PS/2 keyboard driver for input.

This manual describes the features implemented in the current repository. The details were checked against `src/main.rs`, `src/shell.rs`, `src/keyboard.rs`, `src/fs.rs`, `.cargo/config.toml`, and `scripts/build-virtualbox-disk.sh`.

Important: the current file system is stored in RAM only. Files are lost when the virtual machine is restarted, reset, or powered off.

Current Capabilities

The current build includes:

- a keyboard-driven desktop
- a File Manager app
- an Allocation Table app
- text file creation
- text file opening and editing
- file saving through `Ctrl+S`
- file renaming through `Ctrl+R`
- file deletion from the file manager
- cursor movement inside the editor
- Caps Lock and Shift support for text entry
- a simulated indexed-allocation file system

Current Boundaries

The current build does not include:

- persistent disk storage for user-created files
- mouse support
- multitasking
- a graphical window system
- an installer ISO

II. Requirements and Project Setup

Host Tools

Install these tools on the host computer before building the OS:

- Rust and Cargo through `rustup`
- a nightly Rust toolchain
- the `rust-src` component
- the `llvm-tools-preview` component
- `bootimage`
- QEMU tools
- VirtualBox, if the generated VDI will be run in VirtualBox

Check the installed tools with:

```
rustup --version
cargo --version
qemu-img --version
VBoxManage --version
```

Rust Toolchain Setup

Run this setup once from the project root:

```
rustup toolchain install nightly
rustup override set nightly
rustup component add rust-src llvm-tools-preview
cargo install bootimage
```

This project requires nightly Rust because `.cargo/config.toml` uses a custom bare-metal target and `build-std`.

Project Root

Use the project root folder, meaning the folder that contains `Cargo.toml`, `Cargo.lock`, `src/`, `docs/`, and `scripts/`.

Move into that folder before running build commands:

```
cd path/to/xspaceos
```

Replace `path/to/xspaceos` with the actual project location on the computer being used.

III. Build and Boot Procedures

Build the Kernel

To compile the kernel:

```
cargo build
```

To create the bootable raw disk image:

```
cargo bootimage
```

The raw image is created at:

```
target/x86_64-xspaceos/debug/bootimage-xspaceos.bin
```

Build the VirtualBox Disk

To create the VirtualBox disk image, use the helper script:

```
./scripts/build-virtualbox-disk.sh
```

The script runs `cargo bootimage`, then converts the raw image into:

```
xspaceos.vdi
```

Run with QEMU

QEMU is the reference environment for this project.

From the project root, run:

```
cargo run
```

Cargo uses the configured `bootimage runner`, which starts QEMU with the generated boot image.

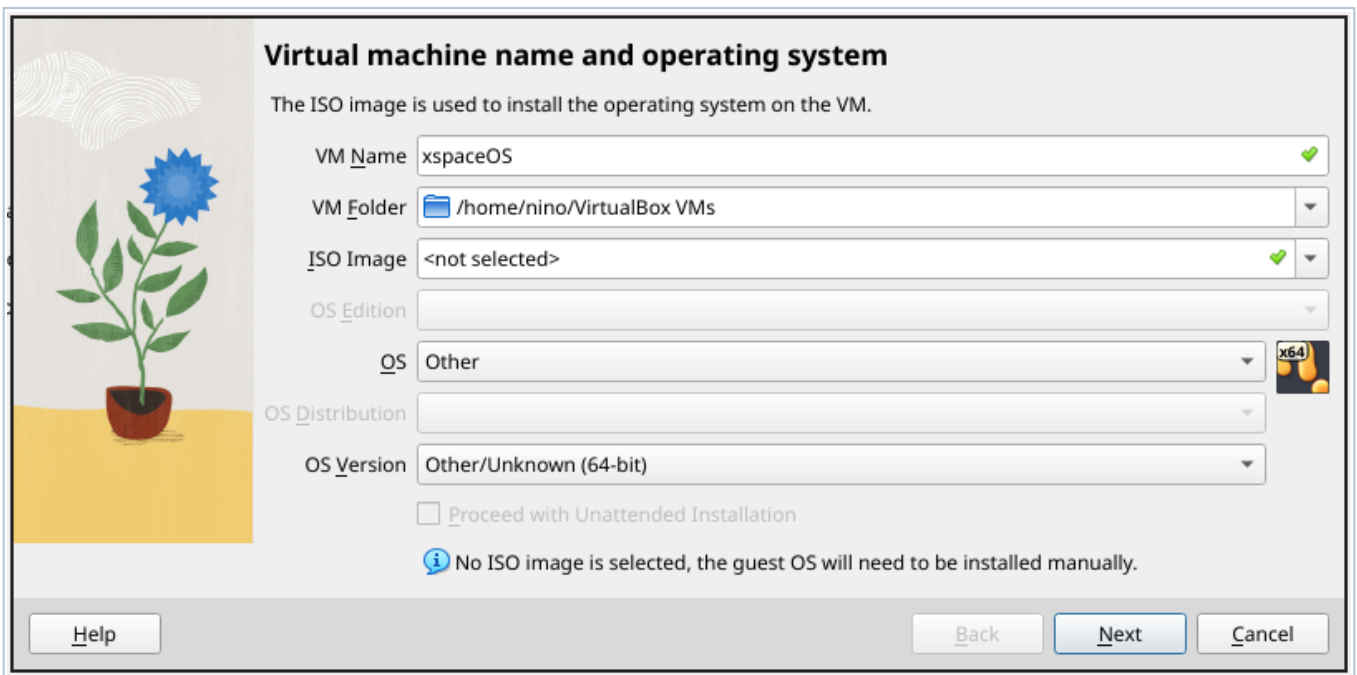
Manual QEMU command:

```
qemu-system-x86_64 -drive format=raw,file=target/x86_64-xspaceos/debug/bootimage-xspaceos.bin
```

Create the VirtualBox Machine

XSpace OS does not use an ISO installer. In VirtualBox, boot from the generated `xspaceos.vdi` disk instead.

1. Open VirtualBox.
2. Click **New**.
3. Use a VM name such as **XSpace OS**.
4. Leave the ISO image field empty.
5. Set **OS** to **Other**.
6. Set **OS Version** to **Other/Unknown (64-bit)**.
7. Continue to the hardware screen.



Virtual machine name and operating system

The ISO image is used to install the operating system on the VM.

VM Name: xspaceOS ✓

VM Folder: /home/nino/VirtualBox VMs

ISO Image: <not selected> ✓

OS Edition:

OS: Other x64

OS Distribution:

OS Version: Other/Unknown (64-bit)

☐ Proceed with Unattended Installation

 No ISO image is selected, the guest OS will need to be installed manually.

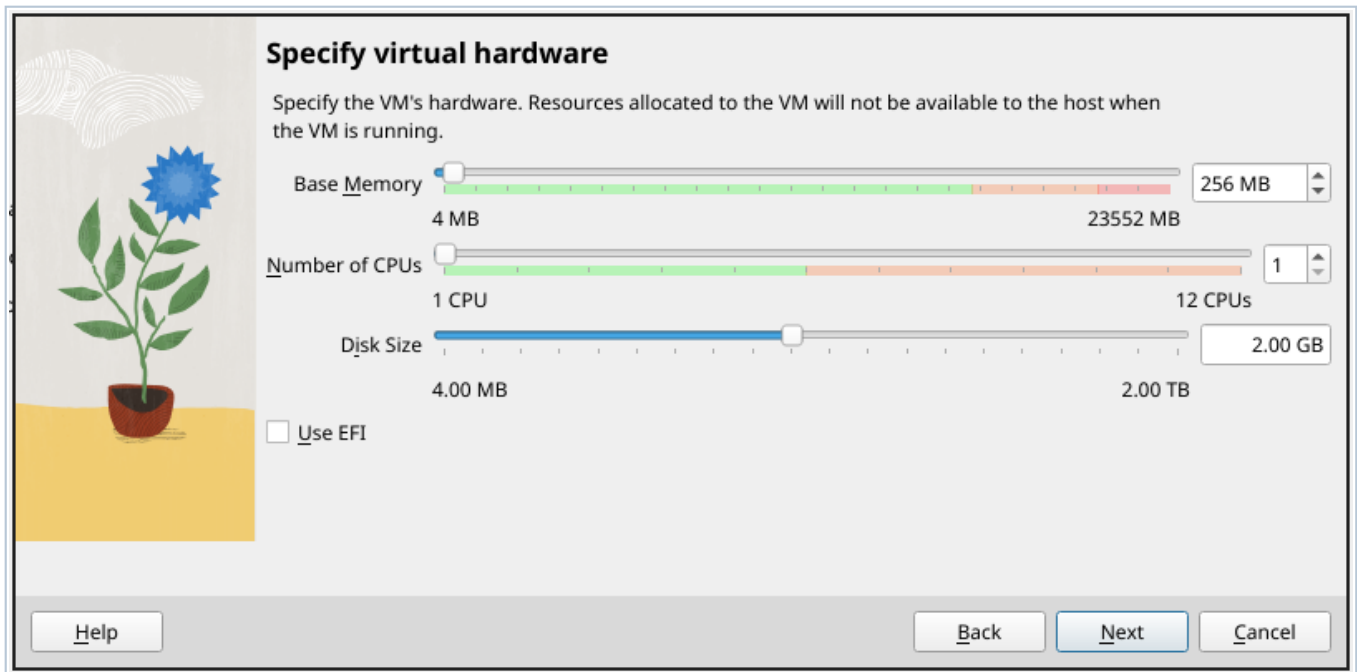
Help **Back** **Next** **Cancel**

Caption: Create the VM without an ISO image and use an **Other/Unknown (64-bit)** guest type.

Configure Virtual Hardware

Recommended VM settings:

- Base Memory: 128 MB minimum; 256 MB is fine
- CPUs: 1
- EFI: disabled
- Boot mode: legacy BIOS



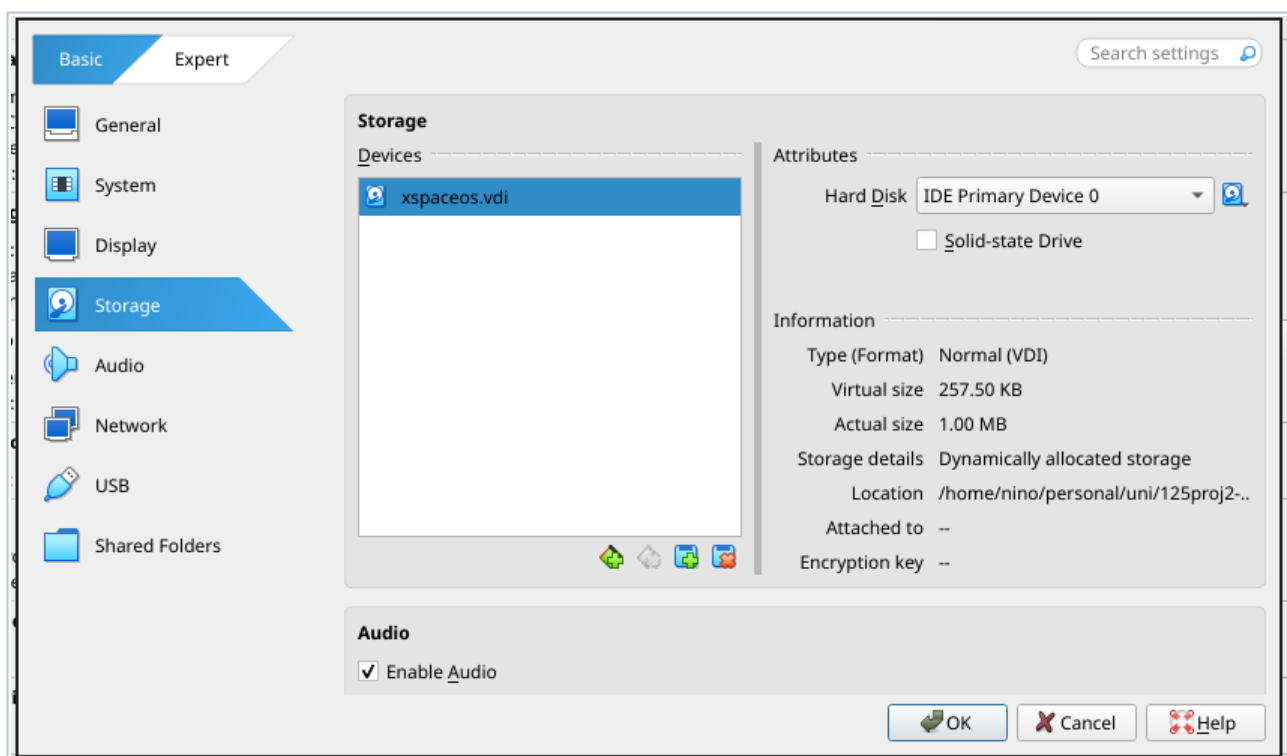
Caption: Keep `Use EFI` unchecked. The current boot image expects legacy BIOS boot.

VirtualBox may ask for a virtual hard disk size during VM creation. It is fine to finish the wizard, because the generated `xspaceos.vdi` will be attached in the next step.

Attach `xspaceos.vdi`

After building the VDI with `./scripts/build-virtualbox-disk.sh` :

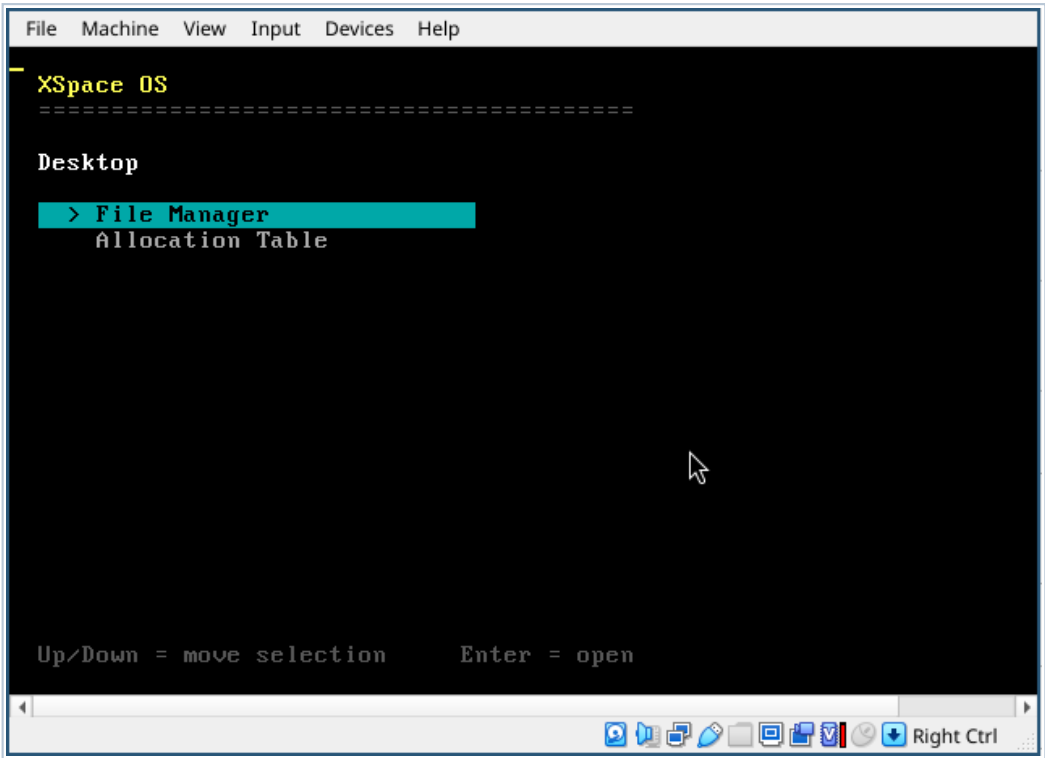
1. Select the VM in VirtualBox.
2. Open `Settings` .
3. Open `Storage` .
4. Remove any temporary blank disk if VirtualBox created one.
5. Attach `xspaceos.vdi` as the VM hard disk.
6. Save the settings.



Caption: Attach the generated xspaceos.vdi file as the VM's hard disk.

Boot the OS

Start the VM. A successful boot shows the XSpace OS desktop.



Caption: The desktop shows the File Manager and Allocation Table apps.

IV. Desktop and Application Navigation

Desktop Screen

The desktop is the first screen after boot.

Desktop controls:

Key	Action
Up / Down	Move the highlighted selection.
Enter	Open the selected app.

Available apps:

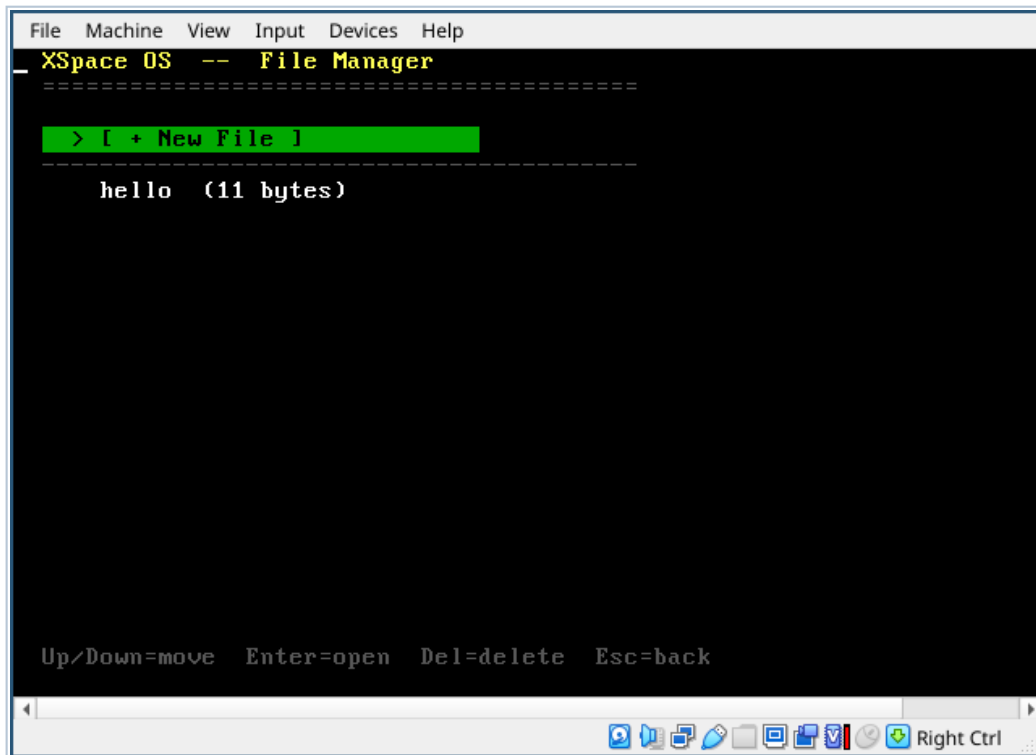
App	Purpose
File Manager	Create, open, edit, rename, and delete files.
Allocation Table	Inspect simulated file allocation blocks.

File Manager Screen

Open File Manager from the desktop by leaving it highlighted and pressing Enter .

The file manager shows:

- [+ New File] at the top
- existing files below it
- each file's current size in bytes



Caption: The file manager lists files and shows each file's byte count.

File manager controls:

Key	Action
Up / Down	Move the highlighted selection.
Enter on [+ New File]	Start creating a new file.
Enter on a file	Open that file in the editor.
Delete on a file	Delete the selected file immediately.
Esc	Return to the desktop.

There is no confirmation prompt before deleting a file. Because the file system is currently in memory only, deleted files cannot be recovered.

V. File Editing Workflow

Create a File

1. Open File Manager .
2. Highlight [+ New File] .
3. Press Enter .
4. Type the file name at the File name: prompt.
5. Press Enter .

If the name is accepted, the editor opens automatically for the new file.

File name rules:

- the name cannot be empty
- the name cannot be longer than 32 bytes
- the name cannot match an existing file

The OS reports an error if the file table is full or the name already exists. If an empty name is entered, the OS returns to the file manager without creating a file.

Open a File

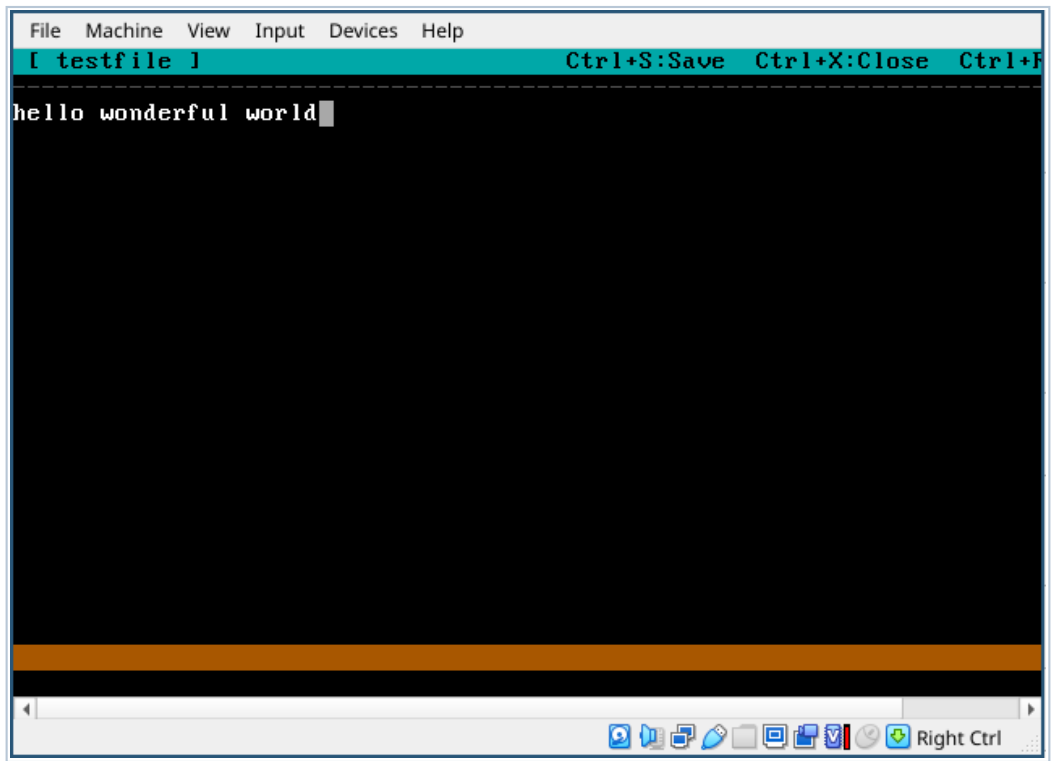
- 1. Open File Manager .
- 2. Highlight the file name.
- 3. Press Enter .

There is no separate read-only viewer. Files are opened in the editor.

Edit Text

The editor is a full-screen text editor. It shows:

- the current file name in the top bar
- editor shortcuts in the top bar
- file contents in the main area
- saved/unsaved status in the bottom bar
- current size out of the 512 byte limit
- CAPS in the bottom bar when Caps Lock is enabled



Caption: The editor supports typing, cursor movement, deletion, saving, closing, and renaming.

Editor controls:

Key	Action
Printable keys	Insert text at the cursor.
Enter	Insert a newline.
Backspace	Delete the character before the cursor.
Delete	Delete the character at the cursor.
Left / Right	Move the cursor one character left or right.
Up / Down	Move through visual editor rows.

Home	Move to the start of the current visual row.
End	Move to the end of the current visual row.
PageUp / PageDown	Move up or down by one visible editor page.
Ctrl+S	Save the file.
Ctrl+R	Rename the file.
Ctrl+X	Close the editor and return to the file manager.

Save before closing. **Ctrl+X** does not ask whether unsaved changes should be kept.

Save a File

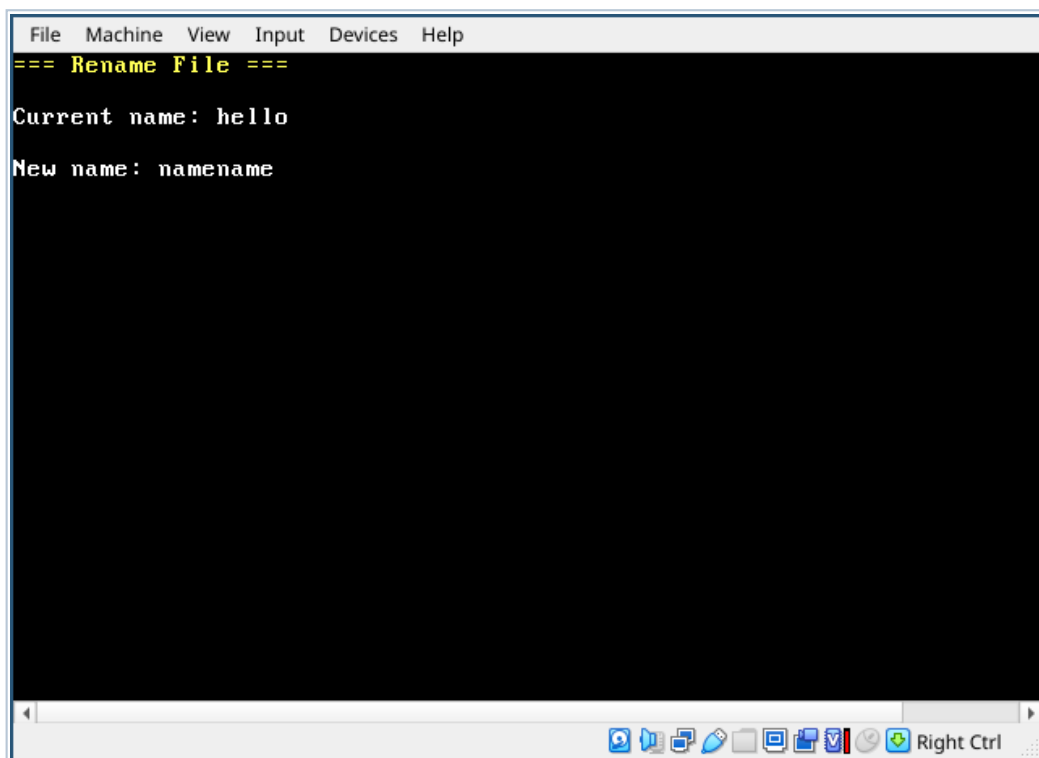
Press **Ctrl+S** inside the editor.

When saving succeeds, the bottom status bar changes to **Saved**. Editing after that changes the status back to **Unsaved**.

The editor stops accepting more text after the file reaches **512** bytes.

Rename a File

1. Open the file in the editor.
2. Press **Ctrl+R**.
3. Type the new name at the **New name:** prompt.
4. Press **Enter**.



Caption: Renaming is done from inside the editor.

Rename rules:

- the new name cannot be empty
- the new name cannot be longer than **32** bytes
- the new name cannot match an existing file

Renaming changes only the file name. It does not change the file contents or the file's allocated data blocks.

Delete a File

- 1. Open File Manager .
- 2. Highlight the file.
- 3. Press Delete .

The selected file is removed immediately. Its index block and data blocks are returned to the free block pool.

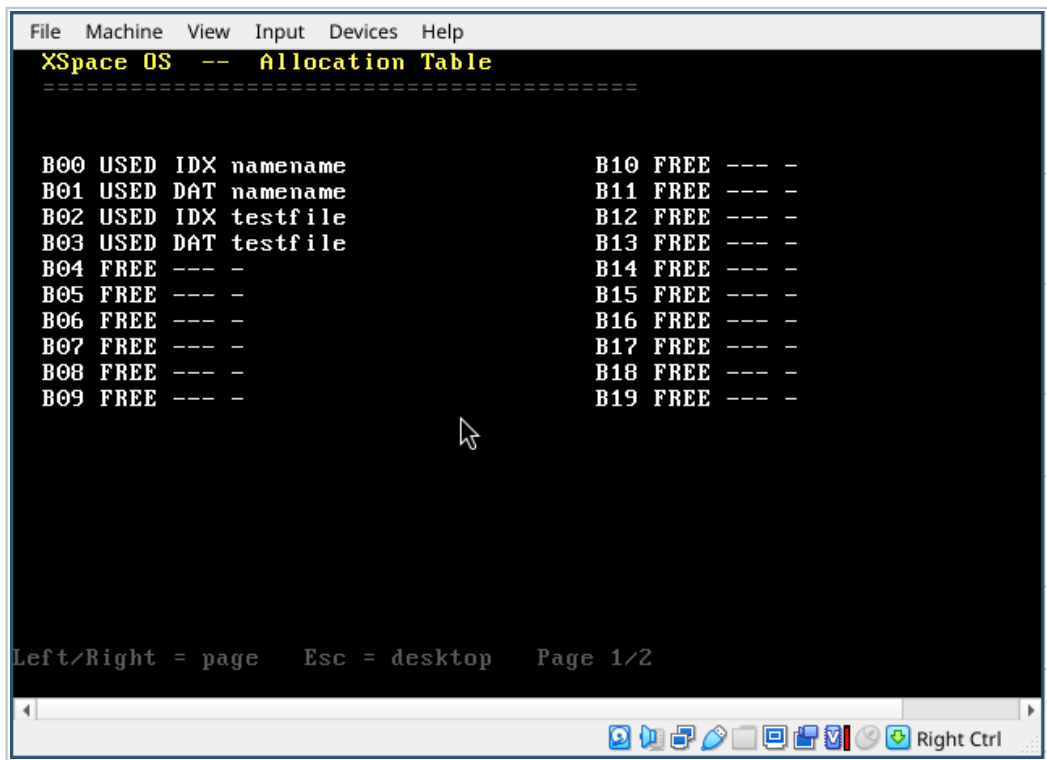
VI. Allocation Table and File System Model

Allocation Table Screen

Open Allocation Table from the desktop to inspect the simulated file allocation state.

The table displays block entries in two columns. Each row contains:

- block number, such as B00
- state: USED or FREE
- block type: IDX , DAT , or ---
- owner file name, if the block is used



Caption: A file uses one IDX block and one or more DAT blocks after content is saved.

Allocation table controls:

Key	Action
Left / Right	Move between allocation-table pages.
Esc	Return to the desktop.

The current file system has 40 simulated blocks, so the allocation table has two pages of 20 blocks each.

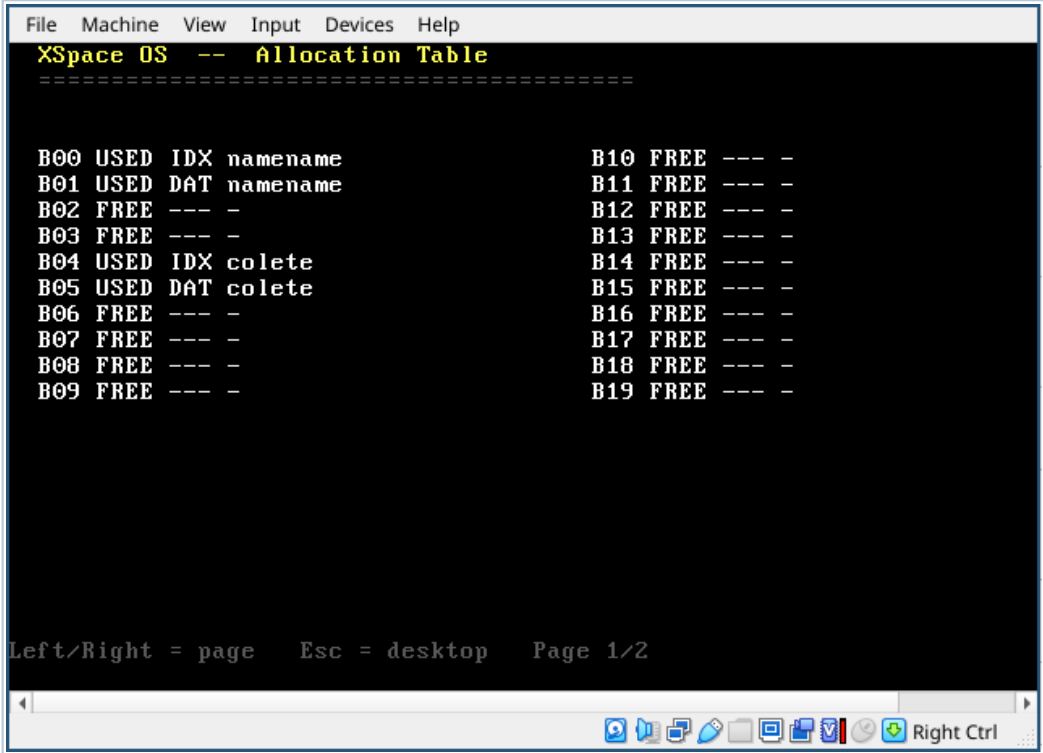
Indexed Allocation Model

XSpace OS currently uses a fixed-size, in-memory file system. It is designed for a small no_std kernel, so it does not require heap allocation.

Storage model:

- files are kept in kernel RAM
- each file reserves one index block when created
- file contents are split into 128 byte data blocks when saved
- each file can own up to four data blocks
- deleting a file frees its index block and data blocks
- saving a file replaces its previous data-block allocation

After deleting a file, its blocks become **FREE** again:



```
File Machine View Input Devices Help
XSpace OS -- Allocation Table
=====

B00 USED IDX namename          B10 FREE --- -
B01 USED DAT namename          B11 FREE --- -
B02 FREE --- -                 B12 FREE --- -
B03 FREE --- -                 B13 FREE --- -
B04 USED IDX colete            B14 FREE --- -
B05 USED DAT colete            B15 FREE --- -
B06 FREE --- -                 B16 FREE --- -
B07 FREE --- -                 B17 FREE --- -
B08 FREE --- -                 B18 FREE --- -
B09 FREE --- -                 B19 FREE --- -

Left/Right = page  Esc = desktop  Page 1/2
```

File System Limits

Current limits:

Limit	Value
Maximum files	8
Maximum file name length	32 bytes
Maximum file content length	512 bytes
Block size	128 bytes
Maximum data blocks per file	4
Index blocks per file	1
Total simulated blocks	40
Persistence	None; RAM only

What this means for users:

- Empty files still use one index block.
- A saved file with 1 to 128 bytes uses one data block.
- A saved file with 129 to 256 bytes uses two data blocks.

- A saved file with 257 to 384 bytes uses three data blocks.
- A saved file with 385 to 512 bytes uses four data blocks.
- Files disappear after reboot or power off.

VII. Keyboard Reference

Text input supports standard US keyboard scancode set 1 behavior for the keys implemented by the current driver.

Supported behavior:

- letters, numbers, spaces, and common punctuation
- Shift for uppercase letters
- Caps Lock for alphabetic capitalization
- Shift with the number row, such as `!`, `@`, `#`, `$`, `%`, `^`, `&`, `*`, `(`, and `)`
- Shift with punctuation, such as `_`, `+`, `{`, `}`, `|`, `:`, `"`, `~`, `<`, `>`, and `?`
- left and right arrow keys
- Home, End, PageUp, PageDown
- Backspace, Delete, Enter, and Esc
- Ctrl+S, Ctrl+R, and Ctrl+X in the editor

Notes:

- Caps Lock affects letters only.
- Holding Shift while Caps Lock is active types lowercase letters.
- Unsupported key combinations are ignored.
- The OS does not use mouse input.

VIII. Troubleshooting

`cargo build` reports `build-std` or target errors

Make sure nightly Rust is active and the required components are installed:

```
rustup override set nightly
rustup component add rust-src llvm-tools-preview
```

`cargo bootimage` is not found

Install `bootimage`:

```
cargo install bootimage
```

`qemu-img` is not found

Install QEMU tools, then verify:

```
qemu-img --version
```

VirtualBox asks for an ISO

Do not select an ISO. XSpace OS boots from `xspaceos.vdi`.

Use this process:

1. Create the VM without installation media.
2. Finish the VM wizard.

3. Open `Settings > Storage`.
4. Attach `xspaceos.vdi` as the hard disk.

VirtualBox does not boot the OS

Check the following:

- `xspaceos.vdi` exists in the project root
- the VDI was rebuilt after the latest code changes
- the VDI is attached as the VM hard disk
- EFI is disabled
- the VM type is a 64-bit `Other/Unknown` style guest
- the raw image boots with QEMU

If QEMU boots but VirtualBox does not, use QEMU for the demo. QEMU is the reference platform for this project.

Files disappear after restarting

This is expected in the current implementation. The file system is in memory only and does not write user files to the VDI.

IX. Demonstration Flow

Use this short flow for a clean demonstration:

1. Build the VDI with `./scripts/build-virtualbox-disk.sh`.
2. Start the VirtualBox VM.
3. Open `File Manager`.
4. Create a file named `notes.txt`.
5. Type:

```
XSpace OS demo file
Created inside the OS editor
Files are currently RAM-only
```

1. Press `Ctrl+S` to save.
2. Press `Ctrl+R` and rename the file to `manual.txt`.
3. Press `Ctrl+X` to return to the file manager.
4. Confirm the renamed file appears with its byte count.
5. Press `Esc` to return to the desktop.
6. Open `Allocation Table`.
7. Show the `IDX` block and `DAT` block entries for the file.

This demonstrates booting, file management, editing, saving, renaming, and indexed allocation inspection.